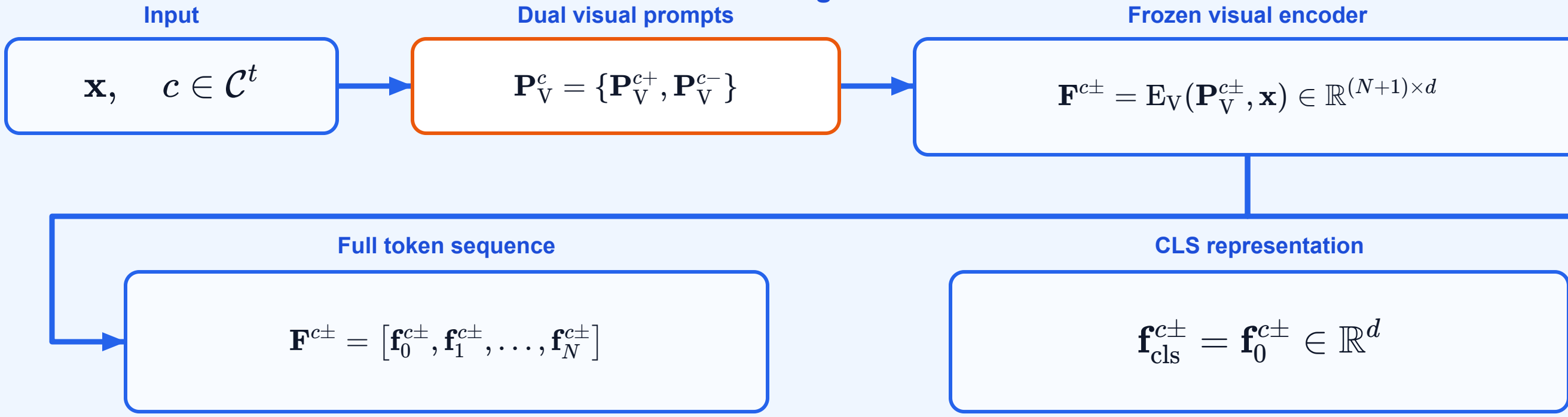


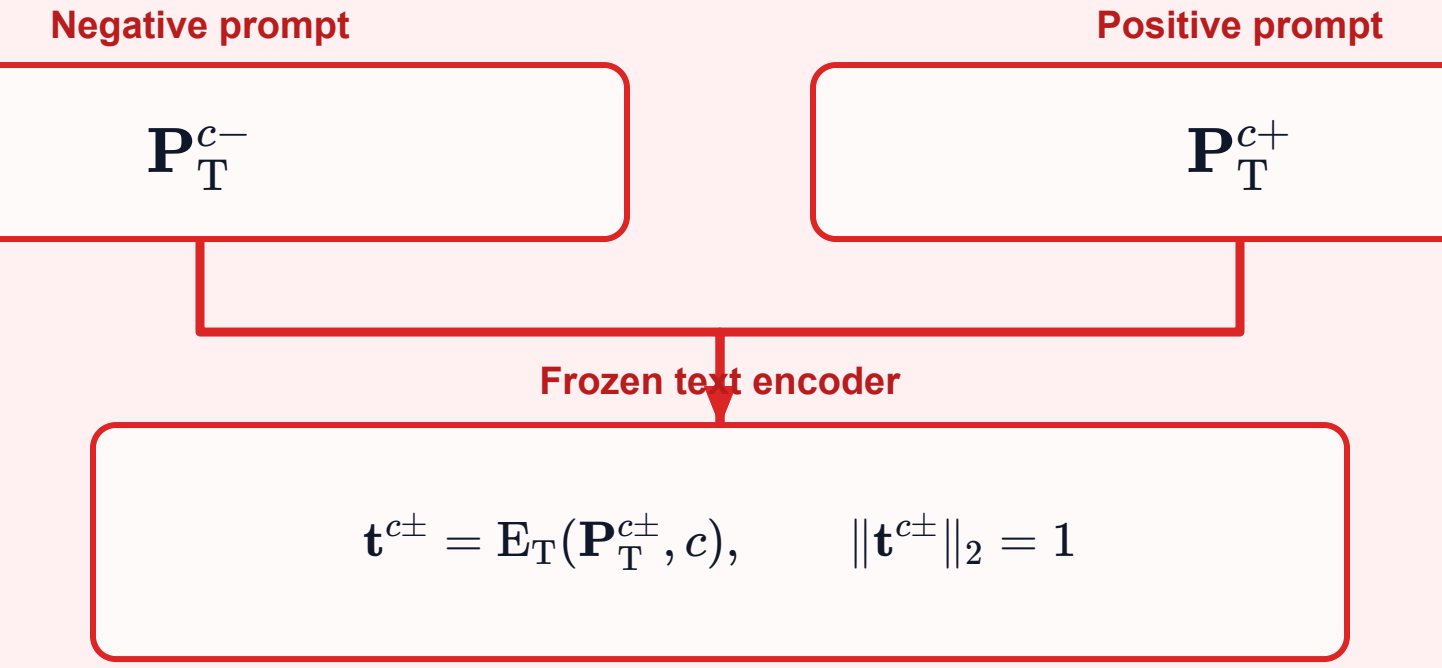
CODE_DDP: EMOTIC B5-C3 Norm-Preserving Feature Correction

The shared CLS Adapter estimates an EMOTIC semantic offset · the offset rotates the DDP pooled representation while preserving its norm · prediction remains the original DDP head

Image branch



Text branch



Explicit representations passed from the shared image/text encoders into DDP pooling, semantic-offset estimation, and the final DDP head

Image → DDP pooling



Text → DDP pooling



Image → CLS Adapter



Text → final DDP head



Original DDP token-attention pooling

Scaled token-text logits

$$q_n^{c\pm} = 20 \langle \mathbf{t}^{c\pm}, \mathbf{f}_n^{c\pm} \rangle$$

Positive-path attention

$$\omega_n^c = \frac{\exp(q_n^{c+})}{\sum_{m=0}^N \exp(q_m^{c+})}$$

Shared-attention pooling

$$\bar{\mathbf{f}}^{c\pm} = \sum_{n=0}^N \omega_n^c \mathbf{f}_n^{c\pm}$$

Class-specific pooled visual representation

$$\bar{\mathbf{f}}^c = (\bar{\mathbf{f}}^{c-}, \bar{\mathbf{f}}^{c+})$$

Shared CLS Adapter: EMOTIC semantic-offset estimation

One frozen mapping shared by all classes and $\pm$ paths

Normalize CLS representation

$$\hat{\mathbf{f}}_{\text{cls}}^{c\pm} = \frac{\mathbf{f}_{\text{cls}}^{c\pm}}{\|\mathbf{f}_{\text{cls}}^{c\pm}\|_2}$$

Estimate EMOTIC domain shift

$$\mathbf{r}^{c\pm} = W_2 \sigma(W_1 \hat{\mathbf{f}}_{\text{cls}}^{c\pm})$$

Semantic feature offset

$$\Delta \mathbf{f}^{c\pm} = \alpha \mathbf{r}^{c\pm}, \quad \alpha = 0.03$$

Norm-Preserving CLS-to-Pooled Feature Correction

$$\dot{\mathbf{f}}^{c\pm} = \|\bar{\mathbf{f}}^{c\pm}\|_2 \frac{\bar{\mathbf{f}}^{c\pm} + \Delta \mathbf{f}^{c\pm}}{\|\bar{\mathbf{f}}^{c\pm} + \Delta \mathbf{f}^{c\pm}\|_2}$$

Original DDP dot-product head applied to the corrected pooled representation

$$\begin{aligned} \tilde{\ell}^{c\pm} &= 100 \langle \mathbf{t}^{c\pm}, \dot{\mathbf{f}}^{c\pm} \rangle \\ &= \ell_{\text{DDP}}^{c\pm} + 100 \langle \mathbf{t}^{c\pm}, \dot{\mathbf{f}}^{c\pm} - \bar{\mathbf{f}}^{c\pm} \rangle, \\ \hat{y}_{c+}^t &= \text{softmax}(\tilde{\ell}^{c-}/\tau(t), \tilde{\ell}^{c+}/\tau(t))_+, \quad c \in \mathcal{C}^{1:t} \end{aligned}$$

New experiment · Norm-Preserving Feature Correction

original DDP token-attention pooling

frozen shared CLS Adapter

feature-space semantic correction

original DDP text-matching head